

HADRIAN

Take Home Test

(CNC Machine Warm Up Program Generator)

Objective

Write a Python script that generates a warmup program for 3-axis CNC machines with different travel limits. The script should allow easy configuration of key warmup parameters and output G-code formatted in Heidenhain Klartext. Optionally, generate code for both Hedenhain and Fanuc controllers.

Machines to Support

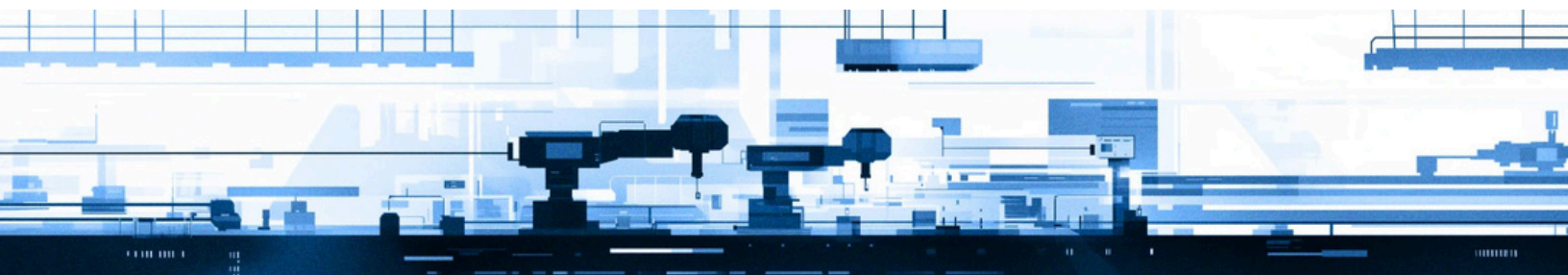
The script should support at least the following three machine configurations:

Machine #	X Travel (mm)	Y Travel (mm)	Z Travel (mm)
1	762	508	500
2	1016	660	500
3	1270	508	500

Requirements / The script should:

1. Generate a valid CNC warmup program for a selected machine.
2. Safely start execution regardless of:
 - Current XYZ position
 - Presets/work offsets
 - Tool offsets
3. Move through the entire XYZ travel limits of the selected machine.
4. Warm up the axis motors/drives gradually by implementing a configurable start and finish feedrate.
5. Warm up the spindle by gradually ramping up from a start RPM, reaching a finish RPM.
6. Optionally activate flood coolant based on user configuration.

The script and G-code should be well commented for readability



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Implementation Details

1. The script should allow the user to define the machine parameters without modification of the python or G-code itself.
2. The generated code should be formatted correctly for execution on a Heidenhain TNC640 controller (Klartext), optionally also generate code for a Fanuc 31i controller.

Example Usage

The candidate should provide an example execution of their script, showing:

- The generated G-code for multiple machine configurations.
- Optional: Generated Fanuc code.

Submission

Submit a Python project (zip file or git repo) that meets the above requirements to the below emails with subject line “Controls Assessment - Full Name”. The script should be well-commented and structured for readability.

Please direct your work to:

- owen@hadrian.co
- jackie.certain@hadrian.co (CC'd)

